Your first EEG research project

A clickable reading guide and first-week plan for a two-student UK A-level team.

The idea: record EEG, label trials, train a classifier, then test a new recording. Nexus supports acquisition and saved-trial classification; continuous device control would be a later stage.

Read in this order

01 **Understand BCI: Brain-Machine Interfaces: Your Brain in Action**

Frontiers for Young Minds. An approachable introduction to how brain signals can become computer commands. Start here.

02 **Understand EEG practically: Setting up for EEG - OpenBCI**

Electrodes, reference connections, alpha rhythms and an eyes-open/closed experiment. Look for sources of movement and electrical noise.

03 **Understand the equipment: Cyton Getting Started Guide - OpenBCI**

How the board, dongle and software fit together. Begin with the diagrams, connection steps and example recordings.

04 **Understand frequencies: Fourier transforms: a visual introduction - 3Blue1Brown**

Why one complicated waveform can contain several rhythms. Focus on the visual explanation before the formal mathematics.

05 **Understand classification: Google: Logistic Regression | Classification**

How features become predictions, what a sigmoid does, and how to read a confusion matrix. Use the introductory explanations first.

06 **Understand a fair test: Data leakage - scikit-learn**

Why training and testing must be separate. Read the concept first; you do not need to understand every Python example.

07 **Understand Nexus: Nexus README | Experiment Lab guide**

The actual recording protocol, labels, quality checks and limitations. Then open Experiment Lab > How the classifier works for the interactive maths lesson.

Optional stretch: Introduction to pyRiemann. Explore covariance matrices and the geometry behind the classifier after the Nexus lesson.

Nexus documentation and the external reading links are public. Blue underlined titles are clickable.

Your first week

Start together, split the reading, then teach it back to each other. Swap practical roles during recording so both understand the full method.

Session	Who and what	A useful output
1 - Foundations	**Both:** read resources 1-2. Discuss the difference between EEG measurement and a BCI.	A shared glossary: electrode, reference, frequency, alpha, artefact and classifier.
2 - Divide the reading	**Student A:** biology, electrodes, hardware and noise (resources 2-3). **Student B:** frequencies, classification and fair testing (resources 4-6).	Each prepares a half-page explanation and three questions for their teammate.
3 - Teach back	**Both:** explain your findings to each other. Work through the Nexus maths lesson without a headset.	Sketch the pipeline: filtered EEG > covariance > tangent-space features > logistic regression.
4 - Software rehearsal	**Both:** use the synthetic board to practise connection, recording, trial labels and exports. Read the Experiment Lab guide.	A rehearsal record of the steps you followed. Synthetic results are not evidence of brain-state classification.
5 - Propose the study	**Both:** agree the question, controls, labels, quality/exclusion rules and separate training/test recordings.	A one-page proposal to discuss with your teacher before participant data collection.

Suggested first research question

Can a classifier distinguish eyes-open from eyes-closed trials in a separate recording from the same participant, better than chance?

Before collecting participant data

- Define what stays constant: participant, electrode positions, reference, task instructions and recording settings.

- Agree how trials will be labelled and which poor-quality trials will be excluded. Record reasons; do not choose exclusions to improve accuracy.

- Keep the final test recording untouched while developing the method. For two equally represented labels, balanced-chance accuracy is 50%.

- Use participant codes, discuss consent and the plan with your teacher, and follow the manufacturer's hardware and electrical-safety instructions.

A weak or negative result can still be useful science. Explain the method, uncertainty and limitations. Task labels describe instructions, not verified thoughts, focus or relaxation. Nexus is an educational signal-exploration tool, not a clinical assessment.